

JOIcode Side-by-Side Report · gpt41_mini

Side-by-side GT vs Generated JOIcode report for **gpt41_mini**. This report is rendered from row-level benchmark outputs and is print-friendly for PDF export.

ROWS 3	AVG DET 69.9000	DET PASS 0.0%	GT EXACT 0.0%	AVG PROMPT TOKENS 14696.3
AVG COMPLETION TOKENS 83.0	AVG LLM LATENCY 3.579s	P50 LLM LATENCY 3.420s	PEAK VRAM 0.000 GB	OOM COUNT 0

RUN METADATA

Results Dir	/home/mgjeong/Desktop/llm/JOILang-Server/gpt_mg/version0_15_update20260413/results/gpt41_mini_retrieval_blocks_equivalence_20260508/retrieval_gpt41_mini_cat6
Output Dir	/home/mgjeong/Desktop/llm/JOILang-Server/gpt_mg/version0_15_update20260413/results/gpt41_mini_retrieval_blocks_equivalence_20260508/retrieval_gpt41_mini_cat6/side_by_side_report_gpt41_mini_failures_only
Suite	paper_with_cloud_ref
Genome	/home/mgjeong/Desktop/llm/JOILang-Server/gpt_mg/version0_15_update20260413/genomes/example_genome.json
Created At	2026-05-08T14:30:38.970760
Categories	6
Limit/Category	-
Rows	151, 152, 153, 154, 155, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169, 170, 171, 172, 173, 174, 175, 176, 177, 178, 179, 180

TOP FAILURE REASONS

- gt_mismatch
- gt_receiver_coverage
- gt_service_coverage
- extraneous
- enum_grounding
- precondition

MODEL RUNTIME SNAPSHOT

MODEL GPT-4.1-mini	RESOLVED MODEL -	WORKER PYTHON -	SERVICE LIST service_list_ver2.0.1.json	CANDIDATE K 1
REPAIR ATTEMPTS 0	DET PROFILE openai			

ROW DETAILS

Row 157 · Category 6

If the weather is rainy and no one is being detected, lock the door and close the valve.

비가 오고 아무도 없는 상태면, 도어락을 잠그고 밸브를 차단해줘.

GT name: IfWeatherIsRainyNoOnelsBeing				
Service list: service_list_ver2.0.1.json				
Output name: LockDoorAndCloseValveIfRainAndNoPresence				
Retrieval cats: PresenceSensor, RainSensor, Door, Valve, DoorLock, WeatherProvider, Pump, Safe, ArmRobot, WindowCovering				
DET 69.9000	PASS No	GT EXACT No	SIMILARITY 0.8457	GT SERVICE COV. 0.7500
RECEIVER COV. 0.7500	DATAFLOW 1.0000	NUMERIC 1.0000	ENUM 1.0000	PROMPT TOKENS 17529
COMPLETION TOKENS 70	TOTAL TOKENS 17599	LLM LATENCY 3.420s	PIPELINE 4.405s	TOK/S 5146.3579
PEAK VRAM 0.000 GB	OOM No	SNIPPET SOURCE service_retrieval_fallback	DEVICE COUNT 10	RETRIEVAL used
CONFIGURED TOP-K 10	EFFECTIVE TOP-K 10			

PROMPT CONTEXT SNAPSHOT

Retrieval premapping was used for this row. Requested top-10 and injected 10 category group(s) into the prompt.

CONTEXT MODE retrieval premapping	SNIPPET SOURCE service_retrieval_fallba ck	RETRIEVAL used	CONFIGURED TOP-K 10	EFFECTIVE TOP-K 10	INJECTED CATEGORIES 10
--------------------------------------	--	-------------------	------------------------	-----------------------	---------------------------

PresenceSensor

1 services · 1 values · 0 functions

VALUES

PresenceSensor_Presence

raw Presence · arg - · ret BOOL · The ability to see the current status of a presence sensor device

FUNCTIONS

None

RainSensor

1 services · 1 values · 0 functions

VALUES

RainSensor_Rain

raw Rain · arg - · ret BOOL · A Device that senses rain

FUNCTIONS

None

Door

RETRIEVED · RANK 3

3 services · 1 values · 2 functions

VALUES

Door_DoorState

raw DoorState · arg - · ret ENUM · Allow for the control of a door

FUNCTIONS

Door_Close

raw Close · arg - · ret VOID · Allow for the control of a door

Door_Open

raw Open · arg - · ret VOID · Allow for the control of a door

Valve

RETRIEVED · RANK 4

3 services · 1 values · 2 functions

VALUES

Valve_ValveState

raw ValveState · arg - · ret BOOL · Controls a valve to open or close it

FUNCTIONS

Valve_Close

raw Close · arg - · ret VOID · Controls a valve to open or close it

Valve_Open

raw Open · arg - · ret VOID · Controls a valve to open or close it

DoorLock

RETRIEVED · RANK 5

3 services · 1 values · 2 functions

VALUES

DoorLock_DoorLockState

raw DoorLockState · arg - · ret ENUM · Allow for the control of a door lock

FUNCTIONS

DoorLock_Lock

raw Lock · arg - · ret VOID · Allow for the control of a door lock

DoorLock_Unlock

raw Unlock · arg - · ret VOID · Allow for the control of a door lock

WeatherProvider

RETRIEVED · RANK 6

7 services · 6 values · 1 functions

VALUES

WeatherProvider_HumidityWeather

raw HumidityWeather · arg - · ret DOUBLE · Provides weather information

WeatherProvider_Pm10Weather

raw Pm10Weather · arg - · ret DOUBLE · Provides weather information

WeatherProvider_Pm25Weather

raw Pm25Weather · arg - · ret DOUBLE · Provides weather information

WeatherProvider_PressureWeather

raw PressureWeather · arg - · ret DOUBLE · Provides weather information

WeatherProvider_TemperatureWeather

raw TemperatureWeather · arg - · ret DOUBLE · Provides weather information

WeatherProvider_Weather

raw Weather · arg - · ret ENUM · Provides weather information

FUNCTIONS

WeatherProvider_GetWeatherInfo

raw GetWeatherInfo · arg DOUBLE | DOUBLE · ret STRING · Provides weather information

Pump

RETRIEVED · RANK 7

2 services · 1 values · 1 functions

VALUES

Pump_PumpMode

raw PumpMode · arg - · ret ENUM · Allows for the control of a pump device

FUNCTIONS

Pump_SetPumpMode

raw SetPumpMode · arg ENUM · ret VOID · Allows for the control of a pump device

Safe

RETRIEVED · RANK 8

3 services · 1 values · 2 functions

VALUES

Safe_SafeState

raw SafeState · arg - · ret ENUM · Allows for the control of the Safe

FUNCTIONS

Safe_Lock

raw Lock · arg - · ret VOID · Allows for the control of the Safe

Safe_Unlock

raw Unlock · arg - · ret VOID · Allows for the control of the Safe

ArmRobot

RETRIEVED · RANK 9

5 services · 2 values · 3 functions

VALUES

ArmRobot_ArmRobotType

raw ArmRobotType · arg - · ret ENUM · Allows for the control of the arm robot

ArmRobot_CurrentPosition

raw CurrentPosition · arg - · ret STRING · Allows for the control of the arm robot

FUNCTIONS

ArmRobot_Hello

raw Hello · arg - · ret VOID · Allows for the control of the arm robot

ArmRobot_SendCommand

raw SendCommand · arg ENUM · ret VOID · Allows for the control of the arm robot

ArmRobot_SetPosition

raw SetPosition · arg ENUM · ret VOID · Allows for the control of the arm robot

WindowCovering

RETRIEVED · RANK 10

6 services · 2 values · 4 functions

VALUES

WindowCovering_CurrentPosition

raw CurrentPosition · arg - · ret INTEGER · Controls a window covering to open or close it

WindowCovering_WindowCoveringType

raw WindowCoveringType · arg - · ret ENUM · Controls a window covering to open or close it

FUNCTIONS

WindowCovering_DownOrClose

raw DownOrClose · arg - · ret VOID · Controls a window covering to open or close it

WindowCovering_SetLevel

raw SetLevel · arg INTEGER · ret INTEGER · Controls a window covering to open or close it

WindowCovering_Stop

raw Stop · arg - · ret VOID · Controls a window covering to open or close it

WindowCovering_UpOrOpen

raw UpOrOpen · arg - · ret VOID · Controls a window covering to open or close it

GT JOICODE

cron="" · period=0

```
if ((#WeatherProvider).weatherprovider_weather == "rain" and
(#PresenceSensor).presencesensor_presence == false) {

    (#DoorLock).doorlock_lock()

    (#Valve).valve_close()

}
```

GENERATED JOICODE

cron="" · period=0

```
if ((#RainSensor).rainsensor_rain == true and (#PresenceSensor).presencesensor_presence ==
false) {
    (#DoorLock).doorlock_lock()
    (#Valve).valve_close()
}
```

FAILURE REASONS

- extraneous
- gt_mismatch
- gt_receiver_coverage
- gt_service_coverage

LINE DIFF

GT	Generated
if ((#WeatherProvider).weatherprovider_weather == "rain" and (#PresenceSensor).presencesensor_presence == false) {	if ((#RainSensor).rainsensor_rain == true and (#PresenceSensor).presencesensor_presence == false) {
	(#DoorLock).doorlock_lock()
(#DoorLock).doorlock_lock()	(#Valve).valve_close()
	}
(#Valve).valve_close()	
}	

Row 175 · Category 6

If the server room temperature is 30 degrees or higher and the AC is off, turn it on and sound the main emergency siren.

서버실 온도가 30도 이상이고 에어컨이 꺼져 있으면, 에어컨을 냉방 모드로 켜고 메인 사이렌을 긴급 모드로 울려줘.

GT name: IfServerRoomTemperatures30DegreesOrHigher				
Service list: service_list_ver2.0.1.json				
Output name: ServerRoomACOnAndEmergencySiren				
Retrieval cats: not used (connected_devices scope)				
DET 69.9000	PASS No	GT EXACT No	SIMILARITY 0.8263	GT SERVICE COV. 0.7500
RECEIVER COV. 0.7500	DATAFLOW 1.0000	NUMERIC 1.0000	ENUM 0.5000	PROMPT TOKENS 12864
COMPLETION TOKENS 76	TOTAL TOKENS 12940	LLM LATENCY 4.835s	PIPELINE 4.845s	TOK/S 2676.0971
PEAK VRAM 0.000 GB	OOM No	SNIPPET SOURCE connected_devices_only	DEVICE COUNT 3	RETRIEVAL skipped_connected_devices_present
CONFIGURED TOP-K 10	EFFECTIVE TOP-K -			

PROMPT CONTEXT SNAPSHOT

Prompt context came from connected_devices, so retrieval was skipped. 4 connected-device category group(s) were injected.

CONTEXT MODE connected_devices scope	SNIPPET SOURCE connected_devices_only	RETRIEVAL skipped_connected_devices_present	CONFIGURED TOP-K 10	EFFECTIVE TOP-K -	INJECTED CATEGORIES 4
---	--	--	------------------------	----------------------	--------------------------

TemperatureSensor

1 services · 1 values · 0 functions

VALUES

TemperatureSensor_Temperature

raw Temperature · arg - · ret DOUBLE · Get the temperature from a Device that reports current temperature

FUNCTIONS

None

CONNECTED

AirConditioner

CONNECTED

4 services · 2 values · 2 functions

VALUES

AirConditioner_AirConditionerMode

raw AirConditionerMode · arg - · ret ENUM · Controls air conditioner mode and temperature settings

AirConditioner_TargetTemperature

raw TargetTemperature · arg - · ret DOUBLE · Controls air conditioner mode and temperature settings

FUNCTIONS

AirConditioner_SetAirConditionerMode

raw SetAirConditionerMode · arg ENUM · ret VOID · Controls air conditioner mode and temperature settings

AirConditioner_SetTargetTemperature

raw SetTargetTemperature · arg DOUBLE · ret VOID · Controls air conditioner mode and temperature settings

Switch

CONNECTED

4 services · 1 values · 3 functions

VALUES

Switch_Switch

raw Switch · arg - · ret BOOL · Allows for the control of a Switch device

FUNCTIONS

Switch_Off

raw Off · arg - · ret VOID · Allows for the control of a Switch device

Switch_On

raw On · arg - · ret VOID · Allows for the control of a Switch device

Switch_Toggle

raw Toggle · arg - · ret BOOL · Allows for the control of a Switch device

Siren

CONNECTED

2 services · 1 values · 1 functions

VALUES

Siren_SirenMode

raw SirenMode · arg - · ret ENUM · Allows for the control of the Siren

FUNCTIONS

Siren_SetSirenMode

raw SetSirenMode · arg ENUM · ret VOID · Allows for the control of the Siren

GT JOICODE

cron="" · period=0

```
if ((#ServerRoom #TemperatureSensor).temperaturesensor_temperature >= 30 and (#ServerRoom
#AirConditioner).switch_switch == false) {

    (#ServerRoom #AirConditioner).airconditioner_setairconditionermode("cool")
}
```

GENERATED JOICODE

cron="" · period=0

```
wait until ((#ServerRoom #TemperatureSensor).temperaturesensor_temperature >= 30 and
(#ServerRoom #Switch).switch_switch == false)
(#ServerRoom #Switch).switch_on()
(#Main #Siren).siren_setsirenmode("emergency")
```

```
(#Main #Siren).siren_setsirenmode("emergency")

}
```

FAILURE REASONS

- enum_grounding
- extraneous
- gt_mismatch
- gt_receiver_coverage
- gt_service_coverage
- precondition

LINE DIFF

GT	Generated
if ((#ServerRoom #TemperatureSensor).temperaturesensor_temperature >= 30 and (#ServerRoom #AirConditioner).switch_switch == false) {	wait until ((#ServerRoom #TemperatureSensor).temperaturesensor_temperature >= 30 and (#ServerRoom #Switch).switch_switch == false)
	(#ServerRoom #Switch).switch_on()
(#ServerRoom #AirConditioner).airconditioner_setairconditionermode("co ol")	(#Main #Siren).siren_setsirenmode("emergency")
(#Main #Siren).siren_setsirenmode("emergency")	
}	

Row 178 · Category 6

If the warehouse fine dust level is 100 or higher and the air purifier is in auto mode, switch it to high mode and say through the warehouse speaker "Switching to high mode".

창고 미세먼지가 100 이상이고 공기청정기가 자동 모드이면, 모드를 강풍 모드로 바꾸고 창고 스피커로 "강풍 모드로 전환합니다"라고 말해줘.

GT name: IfWarehouseFineDustLevels100OrHigher				
Service list: service_list_ver2.0.1.json				
Output name: SwitchWarehouseAirPurifierToHighMode				
Retrieval cats: not used (connected_devices scope)				
DET 69.9000	PASS No	GT EXACT No	SIMILARITY 0.9339	GT SERVICE COV. 0.7500
RECEIVER COV. 0.7500	DATAFLOW 1.0000	NUMERIC 1.0000	ENUM 1.0000	PROMPT TOKENS 13696
COMPLETION TOKENS 103	TOTAL TOKENS 13799	LLM LATENCY 2.483s	PIPELINE 2.494s	TOK/S 5557.1664
PEAK VRAM 0.000 GB	OOM No	SNIPPET SOURCE connected_devices_only	DEVICE COUNT 3	RETRIEVAL skipped_connected_devices_present
CONFIGURED TOP-K 10	EFFECTIVE TOP-K -			

PROMPT CONTEXT SNAPSHOT

Prompt context came from connected_devices, so retrieval was skipped. 3 connected-device category group(s) were injected.

CONTEXT MODE connected_devices scope	SNIPPET SOURCE connected_devices_only	RETRIEVAL skipped_connected_devices_present	CONFIGURED TOP-K 10	EFFECTIVE TOP-K -	INJECTED CATEGORIES 3
---	--	--	------------------------	----------------------	--------------------------

AirQualitySensor

7 services · 7 values · 0 functions

VALUES

AirQualitySensor_CarbonDioxide

raw CarbonDioxide · arg - · ret DOUBLE · Air quality detector device for comprehensive air quality monitoring

AirQualitySensor_DustLevel

raw DustLevel · arg - · ret DOUBLE · Air quality detector device for comprehensive air quality monitoring

AirQualitySensor_FineDustLevel

raw FineDustLevel · arg - · ret DOUBLE · Air quality detector device for comprehensive air quality monitoring

CONNECTED

AirQualitySensor_Humidity

raw Humidity · arg - · ret DOUBLE · Air quality detector device for comprehensive air quality monitoring

AirQualitySensor_Temperature

raw Temperature · arg - · ret DOUBLE · Air quality detector device for comprehensive air quality monitoring

AirQualitySensor_TvocLevel

raw TvocLevel · arg - · ret DOUBLE · Air quality detector device for comprehensive air quality monitoring

AirQualitySensor_VeryFineDustLevel

raw VeryFineDustLevel · arg - · ret DOUBLE · Air quality detector device for comprehensive air quality monitoring

FUNCTIONS

None

AirPurifier

CONNECTED

2 services · 1 values · 1 functions

VALUES

AirPurifier_AirPurifierMode

raw AirPurifierMode · arg - · ret ENUM · Controls air purifier mode

FUNCTIONS

AirPurifier_SetAirPurifierMode

raw SetAirPurifierMode · arg ENUM · ret VOID · Controls air purifier mode

Speaker

CONNECTED

11 services · 2 values · 9 functions

VALUES

Speaker_PlaybackState

raw PlaybackState · arg - · ret ENUM · Speaker device for audio playback and media control

Speaker_Volume

raw Volume · arg - · ret INTEGER · Speaker device for audio playback and media control

FUNCTIONS

Speaker_FastForward

raw FastForward · arg - · ret VOID · Speaker device for audio playback and media control

Speaker_Pause

raw Pause · arg - · ret VOID · Speaker device for audio playback and media control

Speaker_Play

raw Play · arg STRING · ret VOID · Speaker device for audio playback and media control

Speaker_Rewind

raw Rewind · arg - · ret VOID · Speaker device for audio playback and media control

Speaker_SetVolume

raw SetVolume · arg INTEGER · ret INTEGER · Speaker device for audio playback and media control

Speaker_Speak

raw Speak · arg STRING · ret VOID · Speaker device for audio playback and media control

Speaker_Stop

raw Stop · arg - · ret VOID · Speaker device for audio playback and media control

Speaker_VolumeDown

raw VolumeDown · arg - · ret INTEGER · Speaker device for audio playback and media control

Speaker_VolumeUp

raw VolumeUp · arg - · ret INTEGER · Speaker device for audio playback and media control

GT JOICODE

cron="" · period=0

```
if ((#Warehouse #AirQualitySensor).airqualitysensor_finedustlevel >= 100 and (#Warehouse
#AirPurifier).airpurifier_airpurifiermode == "auto") {

    (#Warehouse #AirPurifier).airpurifier_setairpurifiermode("high")

    (#Warehouse #Speaker).speaker_speak("강풍 모드로 전환합니다")

}
```

GENERATED JOICODE

cron="" · period=0

```
if ((#Warehouse #AirQualitySensor).airqualitysensor_dustlevel >= 100 and (#Warehouse
#AirPurifier).airpurifier_airpurifiermode == "auto") {
    (#Warehouse #AirPurifier).airpurifier_setairpurifiermode("high")
    (#Warehouse #Speaker).speaker_speak("강풍 모드로 전환합니다")
}
```

FAILURE REASONS

- gt_mismatch
- gt_receiver_coverage
- gt_service_coverage

LINE DIFF

GT	Generated
if ((#Warehouse	if ((#Warehouse
#AirQualitySensor).airqualitysensor_finedustlevel >= 100	#AirQualitySensor).airqualitysensor_dustlevel >= 100 and
and (#Warehouse #AirPurifier).airpurifier_airpurifiermode	(#Warehouse #AirPurifier).airpurifier_airpurifiermode ==
== "auto") {	"auto") {
	(#Warehouse
	#AirPurifier).airpurifier_setairpurifiermode("high")
(#Warehouse	(#Warehouse #Speaker).speaker_speak("강풍 모드로 전환합니
#AirPurifier).airpurifier_setairpurifiermode("high")	다")
	}
(#Warehouse #Speaker).speaker_speak("강풍 모드로 전환합니	
다")	
}	